

Meet In The Middle (M.I.T.M)

"Concepts & Qns"



Video-1



codestorywithmik



Cs with MIK



codestorywithMIK

Motivation .



Comfort zones are like legacy system - safe and familiar, but entirely incapable of handling exponential growth.

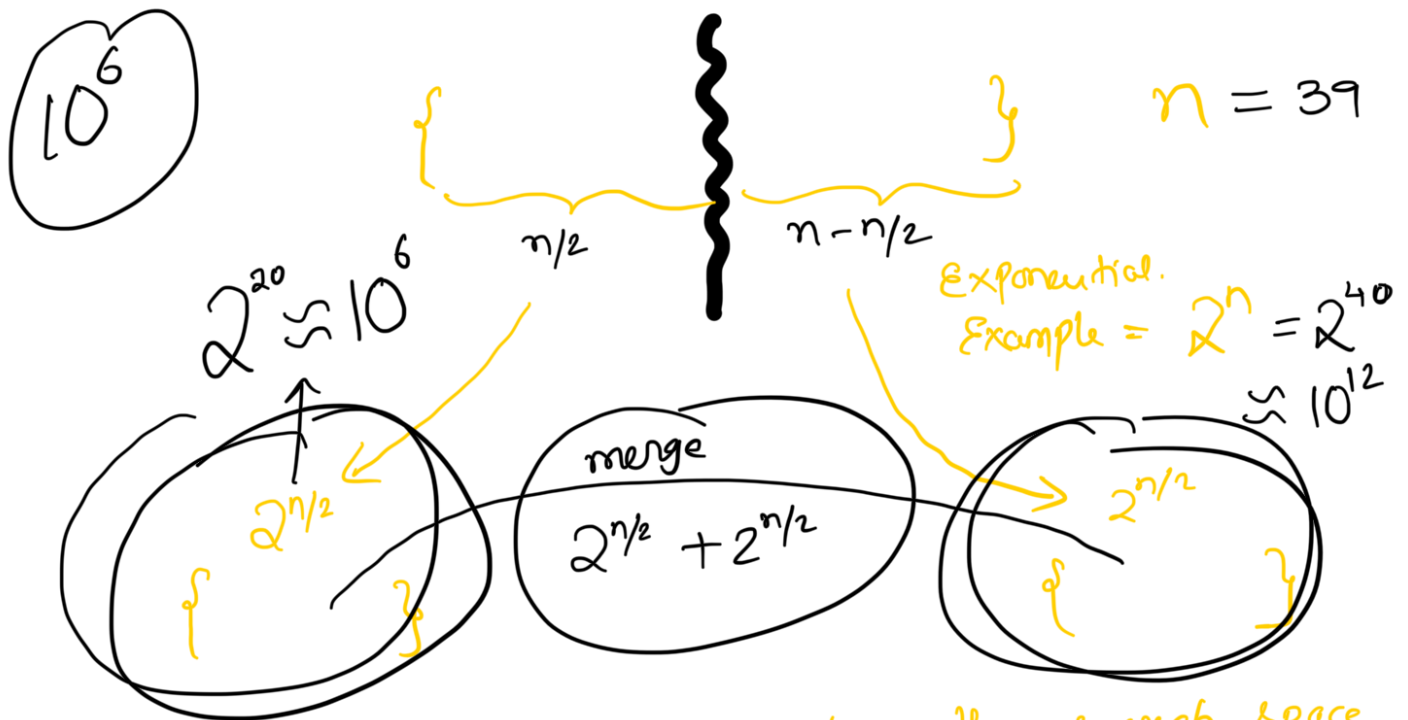
MIK

To truly scale your potential, you must step into the friction of the unknown and constantly stress-test your own limits.

Meet In The Middle (MITM)

What? Why? How?

- Not an algorithm. No defined template.
- It's just a way/technique of solving efficiently.



MITM is a technique where the search space is about equal size.

is divided in two parts of about $\frac{n}{2}$
A separate search is performed for both
the halves and in the end, the results
are combined.

* Used if there is an efficient way to
(merge) Combine the results of two halves.

* The two halves may require less time
than one single search.

Deep Dive...

nums = [1, 2, 3, 2], n=4

x=5

$1 \leq x \leq 10^9$

output = 3

$O(2^n) \approx 10^{12}$

{ 2, 3 }
{ 3, 2 }
{ 1, 2, 2 }

{ 1 }

{ }

skip } DP
take }

$O(2^n)$

$2^{n/2}$
 $2^{20} \approx 10^6$

$2^{n/2}$
 $2^{20} \approx 10^6$

Code using DP:-

Solve(i, x);

Solve(i, x) {

if (i == n) {

if (x == 0) return 1;

return 0;

skip = solve(i+1, x);

if (x >= nums[i])

take = solve(i+1, x - nums[i]);

return skip + take;

dp[n][x]

nums = { 1, 2, 3, 2 }, n=4
n1 = n/2, n2 = n - n1

x=5

$1 \leq n \leq 40$
 $1 \leq x \leq 10^9$

$$\{\} \rightarrow 0$$

$$\{1\} \rightarrow 1$$

$$\{2\} \rightarrow 2$$

$$\{1, 2\} \rightarrow 3$$

$$\text{SubsetSumA} = \{0, 1, 2, 3\}$$

$$x = 5$$

$$2^{n/2}$$

$$2$$

$$3$$

$$5 - 3 = 2$$

$$rem = 5 - 2 = 3$$

$$\{2\} \quad \{3\}$$

$$\{2, 3\} \rightarrow 5$$

$$\{1, 2\} \quad \{2\}$$

$$3 + 2 = x = 5$$

$$\{1, 2, 2\}$$

$$\{\} \rightarrow 0$$

$$\{3\} \rightarrow 3$$

$$\{2\} \rightarrow 2$$

$$\{3, 2\} \rightarrow 5$$

$$\text{SubsetSumB} = \{0, 3, 2, 5\}$$

$$2^{n/2}$$

$$n_1 = n/2$$

$$n_2 = n - n_1$$

SubsetSumA = {0, 1, 2, 3}

$x = 5$

SubsetSumB = {0, 3, 2, 5}

sort
($n_2 \log n_2$) = {0, 2, 3, 5}

$x_{\text{rem}} = x - 2$

= 3

Bin

$\log(n_2)$

$2^{n_1} * \log(n_2)$

→ efficient
u

One more Example Dry Run

nums = {1, 2, 3, 3, 3, 3}, $x = 3$, $n = 6$
 n_1 | n_2 $n/2 =$

{ } → 0

{1} → 1

{2} → 2

{1, 2} → 3

{3} → 3

{1, 3} → 4

{2, 3} → 5

{ } → 0

{3} → 3

{3} → 3

{3} → 3

{3, 3} → 6

{3, 3} → 6

{3, 3} → 6

$$\{1, 2, 3\} \rightarrow 6$$

$$\{3, 3, 3\} \rightarrow 9$$

$$\text{SubsetSumA} = \{0, 1, 2, 3, 3, 4, 5, 6\}$$

$$\text{SubsetSumB} = \{0, 3, 3, 3, 6, 6, 6, 9\}$$

$$x = 3$$

$$0$$

$$\text{rem} = x - 0 = 3 - 0 = 3$$

$$\begin{aligned} \text{low-bound}(3) &= 1 \\ \text{upper-bound}(3) &= 4 \end{aligned}$$

$$\log(n_2)$$

$$4 - 1 = 3$$

$$0 \quad 3$$

$\{ \}$	$\{ 3 \}$	$\{ \}$	$\{ 3 \}$	$\{ 3 \}$	$\{ 3 \}$
$\{ 3 \}$		$\{ 3 \}$		$\{ 3 \}$	

Three subsets -

Story Points ...

$$(i) \quad n_1 = n/2, \quad n_2 = n - n_1$$

$$(ii) \quad \text{SubsetSumA}, \quad \text{SubsetSumB}$$

(.) Sort SubsetSumB

(.) Merge efficiently to find subsets sum == target

↳ Pick from SubsetSumA and Search in SubsetSumB

(x-s)
↓
upperBound
lowerBound

How to find Subset Sums ??

(Recursion / Nested loop)

nums = { 2, 5, 1 } , nl=3

$2^n = 8$

Decimal	mask	subset
0	000 i=2 i=1 i=0	{ }
1	001	{ 2 }
2	010	{ 5 }
3	011	{ 2, 5 }
4	100	{ 1 }
5	101	{ 2, 1 }
6	110	{ 5, 1 }
7	111	{ 2, 5, 1 }

mask = 011

2+5=7

mask
[0, 2^n-1]

```

for (mask = 0; mask < (1 << n); mask++) {
    int sum = 0; // Subset Sum
    for (i = 0; i <= n-1; i++) {
        if (mask & (1 << i)) {
            sum += nums[i];
        }
    }
    SubSetSumA.push_back(sum);
}

```

SubSetSumB

How this is different
from Divide & Conquer
???

D&C

{ 2, 3, 1, 4 }

MITH



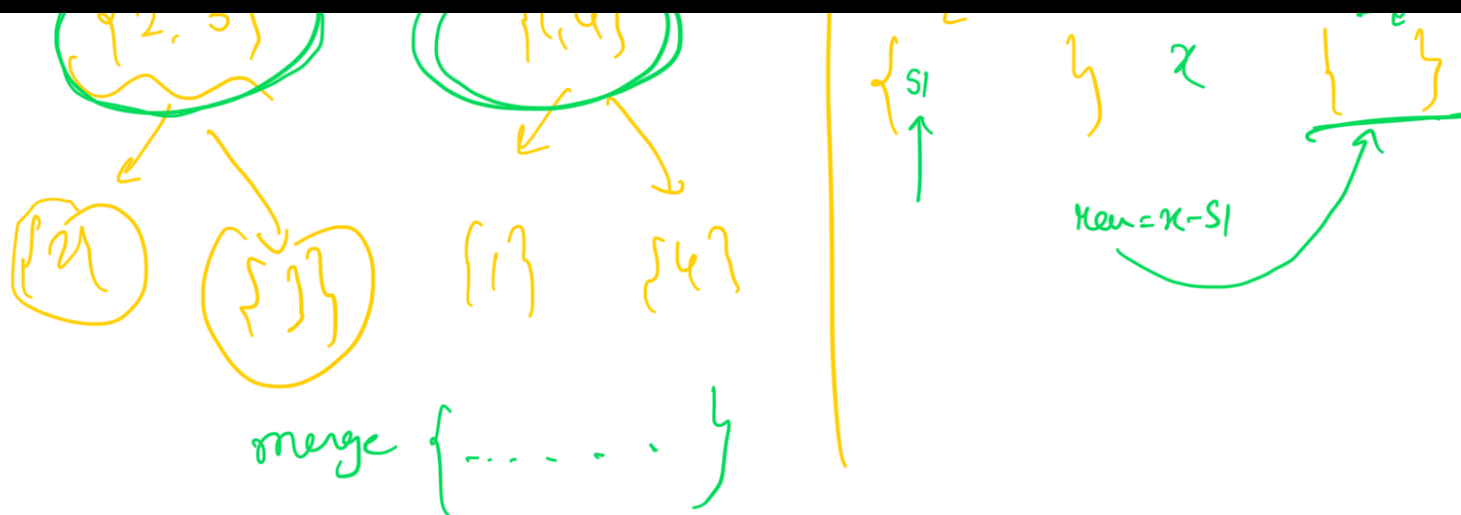
1, 2, 3

1, 2, 3

{

}

Send
Back



When to think of applying
MITM $\rightarrow$???

If natural solution is exponential $\rightarrow$

① Constraints : $n \leq 40$ or (~ 30 to 45)

$\downarrow$
 Too big for brute Force (2^n)

But, perfect for $2^{n/2}$

② Brute force is exponential & not polynomial:-
(2^n) ~~(n^2/n^3)~~

→ Try all 2^n subsets

→ Try all 2^n combinations etc...

If Brute Force is $O(n^2)$ or $O(n^3)$
MITM has nothing to halve.

💡 "Exponential"

③ DP not working because of huge constraints.

④ The result is choose one choice from first part,
with one choice from a second part.

"MITM splits problem clearly in two halves only"

⑤ Matching / Counting / Optimizing over pairs
from two groups.

MITM

